



INTERNATIONAL JOURNAL OF CREATIVE RESEARCH THOUGHTS (IJCRT)

An International Open Access, Peer-reviewed, Refereed Journal

FOREST FIRE DETECTION AND PREVENTION SYSTEM

¹Dinesh J, ²Dinesh Aravind G, ³Hari Krishnan I, ⁴Aruna P

¹UG Scholar, ²UG Scholar, ³UG Scholar, ⁴Assistant Professor

¹Department of Computer Science and Engineering,

¹Sri Ramakrishna Engineering College, Coimbatore, India

Abstract: In the advancing world, it is very crucial to protect our environment. Many incidents of man made and natural disasters were happening around the world. Forest fires are one such catastrophe for environment. Once the fire inside deep forest starts, it burns and destroys everything and spreads everywhere within the forest. Such forest fires disasters should be curbed in order to protect fauna and flora habitats in the forest. The objective of this work is to design and implement an IoT based system which would detect the forest fires and sends the exact location information to the concerned officials which would help fire fighting personnel to extinguish the fire in the location and bring it under control before the fire spreads to the entire forest. The module consists of a fire extinguisher that shuts off a small amount of fire by sensing the temperature of the existing area which would prevent more man power and efforts. This would prevent the fire to spread over a huge area and also able to take precautionary measures in order to prevent the fire which may occur in near future. This model would help in making a decision for fire-extinguishing or fire prevention rightly and real-timely by related government departments.

Index Terms – Internet of Things , Global System for Mobile communication , Global Positioning System .

I. INTRODUCTION

Forest is the major contributor of Earth's ecological balance. Thousands of hectares around the globe are destroyed by the forest fires every year causing tragic loss of fauna and flora. Forest fires are a great menace to ecologically healthy grown forests and protection of the environment. The most destructive fires are usually preceded by extreme high temperatures, low relative humidity, strong winds, lightning which combine to create ideal conditions for the rapid spread of fire. Based on the Forest Survey of India's data on forest fire it is stated that around 50% of the forest areas are fire prone. Based on the forest inventory records, 54.40% of forests in India are exposed to occasional fires, 7.49% to moderately frequent fires and 2.405 to high incidence levels while 35.71% of India's forests have not yet been exposed to fires of any real significance.

Internet of Things:

The Internet of things (IoT) is a system of interrelated computing devices, mechanical and digital machines provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. The definition of the Internet of things has evolved due to the convergence of multiple technologies, real-time analytics, machine learning, commodity sensors, and embedded systems. Traditional fields of embedded systems, wireless sensor networks, control systems, automation including home and building automation, and others all contribute to enabling the Internet of things.

Embedded IoT:

It is essential to know about the embedded devices while learning the IoT or building the projects on IoT. The embedded devices are the objects that build the unique computing system. These systems may or may not connect to the Internet. An embedded device system generally runs as a single application. However, these devices can connect through the internet connection, and able to communicate through other network devices. The embedded system can be of type microcontroller or type microprocessor. Both of these types contain an integrated circuit (IC). The essential component of the embedded system is a RISC family microcontroller like Motorola 68HC11, PIC 16F84, Atmel 8051 and many more. The most important factor that differentiates these microcontrollers with the microprocessor like 8085 is their internal read and writable memory. The essential embedded device components and system architecture are specified below.

II. LITERATURE REVIEW

M.Trinath basu, Ragipati Karthik, J.Mahitha, V.Lokesh Reddy [1], It has been found in a survey that 80% losses caused due to fire would have been kept away from if the fire was identified promptly .Node Mcu based IoT empowered fire indicator and observing framework is the answer for this issue. In this task, we have assembled fire finder utilizing Node Mcu which is interfaced with a temperature sensor, a smoke sensor and signal.

Junguo Zhang, Wenbin Li, Zhongxing Yin, Shengbo Liu, Xiaolin Guo [2], A cluster-based wireless sensor network paradigm for forest fire real-time detection was put forward in this paper. Some key questions were discussed emphatically, such as the ad hoc network related technology, the node hardware designing, the forest-fire forecasting model and the propagation characteristic of UHF wireless signal and so on.

T.Sasikumar, P.Sriramya [3], in their research they developed an Arduino platform based IOT enabled fire detector and monitoring system is the solution to this problem. In this project we have built fire detector using Arduino UNO which is interfaced with a temperature sensor, a smoke sensor and a buzzer.

Ranjith E, Padmabalaji D, Sibisubramanian S, Ms. S Radhika [4], their reasearch uncontrolled blazes occurred by weather, wind, and dry underbrush, can burn acres of land and consume everything in their paths which cause significant damage to natural and human resource. The objective of this work is to design a IoT based system that can detect the fire as early as possible before the fire spread over the large area and to prevent poaching.

Abhinav Kumar Sharma, Md Faiz Raza Ansari, Md Firoz Siddiqui, Mirza Ataullah Baig GLBITM, Gr. Noida, U.P [5], proposed a scheduling whenever it detects fire or smoke then it instantly alerts the user about the fire through the ethernet module. For this purpose, we are using Arduino Uno which is from Arduino family. Also, the Arduino interfacing with LCD display is done to display the status of the system whether the Smoke and Overheat is detected or not.

III. PROPOSED WORK

The proposed system is designed in a way that provides precision information about the fire affected area. The temperature sensor detects the atmospheric temperature and when it exceeds the threshold value the flame detection sensor detects whether the fire is ignited or not and sends the signal to the micro controller. Once the micro controller receives the input, the fire extinguisher starts to extinguish the fire over an area. If there is an increase in the intensity of the fire in the forest, the GPS module attached to the system sends the location to the registered number of the forest officials.

Advantages:

- The proposed system detects the fire more spontaneously and provides the accurate information about the fire before it reaches the majority of the land area.
- The latitude and longitude coordinates of the fire is captured from the satellite and the affected area information is sent to the fire fighting officials so that they can head to the area at the earliest.
- Fire extinguisher slows down the fire at the initial stage, so that the damage to the natural assets will be reduced.

IV. IMPLEMENTATION

3.1 ATMEGA 328 MICRO CONTROLLER

The ATmega328 is a single-chip microcontroller created by Atmel in the mega AVR family, later Microchip Technology acquired Atmel in 2016. It has a modified Harvard architecture 8-bit RISC processor core. The modified Harvard architecture is a variation of the Harvard computer architecture that, unlike the pure Harvard architecture, allows the contents of the instruction memory to be accessed as data. Most modern computers that are documented as Harvard architecture are, in fact, modified Harvard architecture.

3.2 DHT11 TEMPERATURE SENSOR

DHT11 sensor consists of a capacitive humidity sensing element and a thermistor for sensing temperature. The humidity sensing capacitor has two electrodes with a moisture holding substrate as a dielectric between them. Change in the capacitance value occurs with the change in humidity levels. The IC measure, process this changed resistance values and change them into digital form. For measuring temperature this sensor uses a Negative Temperature coefficient thermistor, which causes a decrease in its resistance value with increase in temperature.

3.3 FLAME DETECTION SENSOR

A flame-sensor is one kind of detector which is mainly designed for detecting as well as responding to the occurrence of a fire or flame. The flame detection response can depend on its fitting. It includes an alarm system, a natural gas line, propane & a fire suppression system. This sensor is used in industrial boilers. The main function of this is to give authentication whether the boiler is properly working or not. The response of these sensors is faster as well as more accurate compare with a heat or smoke detector because of its mechanism while detecting the flame. This sensor can be built with an electronic circuit using a receiver like electromagnetic radiation. This sensor uses the infra red flame flash method, which allows the sensor to work through a coating of oil, dust, water vapour, otherwise ice.

3.4 GSM MODEM

GSM modem or GSM module is a hardware device that uses GSM mobile telephone technology to provide a data link to a remote network. GSM modems typically provide TTL-level serial interfaces to their host. They are usually used as part of an embedded system. It can be a dedicated modem device with a serial, USB or Bluetooth connection, or it can be a mobile phone that provides GSM modem capabilities. GSM modems can be a quick and efficient way to get started with SMS, because a special subscription to an SMS service provider is not required. In most parts of the world, GSM modems are a cost effective solution for receiving SMS messages, because the sender is paying for the message delivery.

3.5 GPS MODULE

The GPS module is used for navigation purposes. The module simply checks its location on earth and provides output data which is longitude and latitude of its position. It is from a family of stand-alone GPS receivers featuring the high performance u-blox 6 positioning engine. These flexible and cost effective receivers offer numerous connectivity options in a miniature package. The compact architecture, power and memory options make GPS module ideal for battery operated mobile devices with very strict cost and space constraints. Its Innovative design provides excellent navigation performance even in the most challenging environments. This module is one of popular GPS modules on the market and is also cheap to buy. The location data provided by it is accurate enough to satisfy most applications. And for it to be included in smart phones and tablets design points out its efficiency.

V. RESULTS AND DISCUSSION

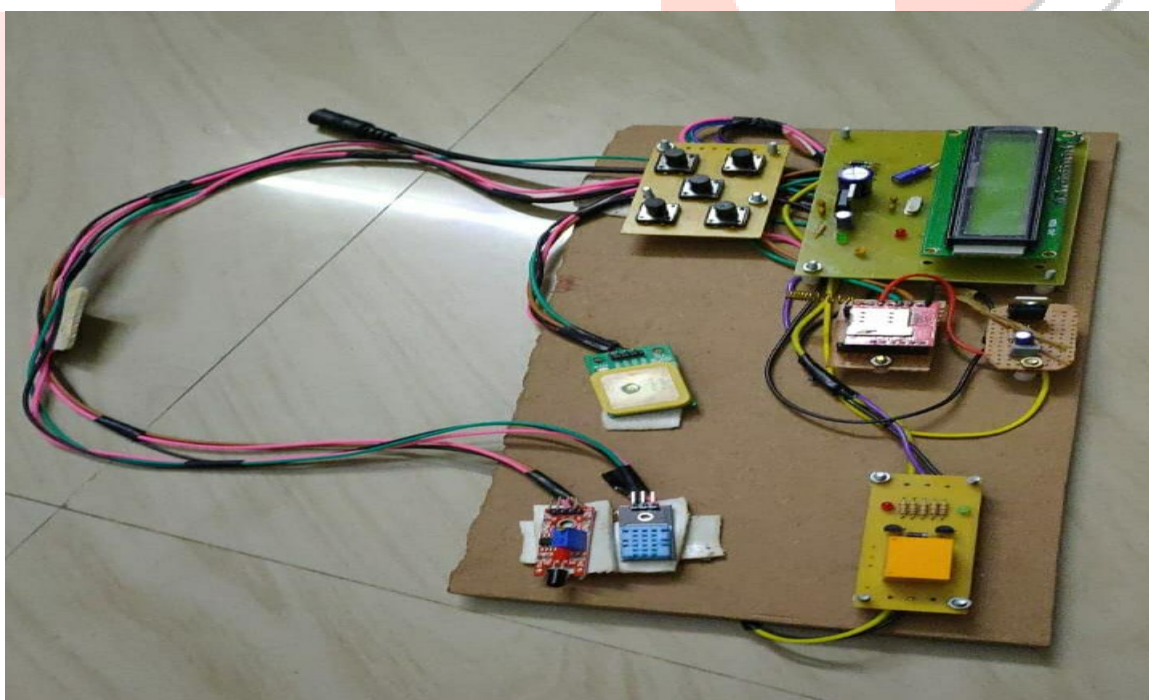


Fig.1 Hardware Components circuit



Fig.2 Fire extinguisher setup coupled with the circuit



Fig.3 Assembled system over a tree



Fig.4 System operating model



Fig.5 Fire extinguisher is set on

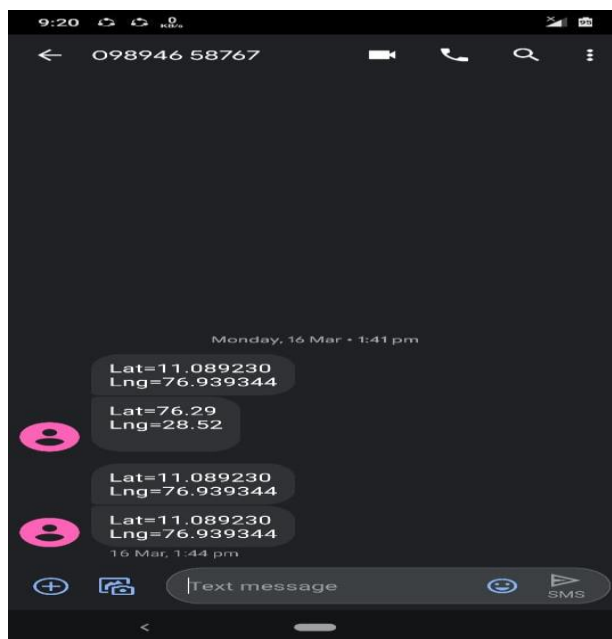


Fig.6 GPS coordinates received from the GSM module

VI. CONCLUSION

It is essential to protect our environment and nature. The main aim of the IoT based system is to detect the forest fire as early as possible and protect our entire system from fire-related calamities. In addition to this, the proposed model is designed to incorporate with the Autonomic features like self- monitoring and self-healing so that the ubiquitous environment that is created for a specific objective can be attained with a robust and fault-tolerant system.

Acknowledgment

We express our deepest gratitude to Project Guide and Coordinator of our Computer Science and Engineering department who gave constant support and encouragement throughout this work.

REFERENCES

- [1] Junguo Zhang, Wenbin Li, Zhongxing Yin, Shengbo Liu, and Xiaolin Guo, "Forest fire detection system based on wireless sensor network", IEEE Conference on Industrial Electronics and Applications, 2009.
- [2] Abhinav Kumar Sharma, Md Faiz Raza Ansari, Md Firoz Siddiqui, and Mirza Ataullah Baig, "IoT enabled forest fire detection and online monitoring system", International Journal of Current Trends in Engineering & Research pp. 50 – 54, 2017.
- [3] M. Trinath Basu, Ragipati Karthik, J. Mahitha, and V. Lokesh Reddy, "IoT based forest fire detection system", International Journal of Engineering & Technology, 2018.
- [4] Ranjith E, Padmabalaji D, Sibisubramanian S, and Ms. Radhika S, "An IoT based forest fire detection system using raspberry pi 3", International Research Journal of Engineering and Technology, 2019.
- [5] T. Saikumar and P. Srirama, "IoT enabled forest fire detection and alerting system", International Journal of Recent Technology and Engineering, 2019.